Homework 4 (Polyhedra)

February 3, 2026

Problems **1)** and **2)** are worth 10 points each. Problems **3)** and **4)** are worth 20 points each.

1)

Determine which of the following sets are polyhedra. If a set is a polyhedron, prove it. If not, explain why not.

a)

$$S = \left\{x \in \mathbb{R} | x^2 - 3x - 10 \leq 0\right\}$$

b)

$$S = \left\{(x, y) \,|\, |y| \geq x^2 + 4x + 1\right\}$$

c)

The empty set:

$$S = \emptyset$$

2)

Let P be a bounded polyhedron in $\mathbb{R}^n$. Let $\vec{a} \in \mathbb{R}^n$ and $\vec{b} \in \mathbb{R}$. Define

$$Q = \{\vec{x} \in P | \vec{a} \cdot \vec{x} = b\}$$

Show that every extreme point in Q is an extreme point of P or a convex combination of two adjacent extreme points of P.

3)

Let P and Q be polyhedra in $\mathbb{R}^n$. Let

$$P + Q = \{\vec{x} + \vec{y} | \vec{x} \in P, \vec{y} \in Q\}$$

Show the following:

a)

$P + Q$ is a polyhedron.

b)

If P and Q are bounded, then $P + Q$ is bounded.

c)

Every extreme point in $P + Q$ is the sum of an extreme point in P and an extreme point in Q.

d)

If P is is a polyhedron, then

$$T(P) = \{T(\vec{x}) \,|\vec{x} \in P\}$$

is a polyhedron for any affine transformation, T.

4)

Let P and Q be polyhedra in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively. We say that P and Q are isomorphic if there exist affine mappings $f : P \to Q$ and $g : Q \to P$ such that $g(f(\vec{x})) = \vec{x}$ for all $\vec{x} \in P$ and such that $f(g(\vec{y})) = \vec{y}$ for all $\vec{y} \in Q$.

a)

If P and Q are isomorphic, show that there exists a bijection between the extreme points of P and Q.

b)

Let $P = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}, \vec{x} \geq 0\right\}$ and let $Q = \left\{(\vec{x}, \vec{z}) \in \mathbb{R}^{n+k} | A\vec{x} - \vec{z} = \vec{b}, \vec{x}, \vec{z} \geq 0\right\}$ for some $A \in \mathbb{R}^{k \times n}$. Show P and Q are isomorphic. (Note, this problem justifies why we are okay to use standard form when working with polyhedra).